UUM
Universiti Utara Malaysia

FINAL EXAMINATION
FIRST SEMESTER 2023/2024 SESSION

COURSE CODE/NAME	:	STIW3064/COMPONENT-BASED DEVELOPMENT
DATE	:	20 FEBRUARY 2024 (TUESDAY)
TIME	:	2.30 P.M. – 5.00 P.M. (2 ½ HOURS)
VENUE	:	BK3 COLGIS

INSTRUCTIONS :

1. This examination paper has **ELEVEN (11)** structured questions in **TEN (10)** printed pages excluding the cover page.
2. Answer **ALL** questions in the space provided.
3. You are **NOT ALLOWED** to remove the examination paper from the examination hall

MATRIC NO.: _____ (in words) | | | | | | | (in numbers)

IDENTIFICATION CARD NO.: | | | | | | | | | | | | |

LECTURER NAME: _____

GROUP: | | **TABLE NO.:** | | | | |

DO NOT OPEN THIS EXAMINATION PAPER UNTIL INSTRUCTED

INSTRUCTION: ANSWER ALL QUESTIONS IN THIS SECTION.

1. Software developers can choose any software development approach that is appropriate for them to apply.

   a) Name and describe **THREE (3)** software development approaches.

   (9 marks)

   b) State **ONE (1)** software process model from any of the software development approaches given in **1(a)**.

   (1 mark)

2. Component-based development promises several advantages that are claimed to counter the issues faced by developers who use other development methods.

a) Describe **TWO (2)** advantages of component-based development.

(6 marks)

b) State **TWO (2)** advantages and their corresponding issues that they counter.

(4 marks)

3. Component-based software development follows the processes stipulated in the software development life-cycle (SDLC) methodology. However, several processes especially exist in component-based development (CBD).

   a) State **ALL** the processes that especially exist in CBD.

   (6 marks)

   b) Explain **ONE (1)** of the process stated in **3(a).**

   (4 marks)

c) Justify the relation between **ONE (1)** of the processes stated in **3(a)** with any of the processes in the software development life-cycle.

(5 marks)

4. Software component development is one of the crucial processes in component-based development.

a) What is a software component?

(2 marks)

b) What is the generic name of a component standard?

(1 mark)

c) List **ALL** the basic elements of a component standard.

(3 marks)

d) Discuss the usage of a component standard in ensuring the reusability of a newly developed component.

(5 marks)

5. Identify **TWO (2)** types of component-based development products that need to be tested.

(4 marks)

6. Reusable components can be obtained from several sources.

a) Elaborate the process of selecting the required components.

(5 marks)

b) Identify the outcome of the process stated in **6(a).**

(2 marks)

c) Explain the process that will be performed on the outcome stated in **6(b).**

(3 marks)

7. To develop software using any component-based development model, the developers need to apply a component model that is deemed appropriate in their development context. Discuss **ONE (1)** reason to justify the appropriateness of the selected component model to be applied in component-based development.

(5 marks)

8. In developing an application using component-based development, emphasis is given to the reuse of existing components. However, a component may not be ready for reuse due to some reasons.

   a) Provide **ONE (1)** reason why a component is not ready for reuse.

(2 marks)

b) Explain **TWO (2)** techniques that could be applied in resolving the problem.

(8 marks)

9. In adapting a component to be reused, the component has to be checked for its compatibility with the context of its reuse. Justify the aim of performing the compatibility check and how it is done.

(5 marks)

10. In developing an application using component-based development, components are normally composed to create an application.

    a) Elaborate the process of component composition and integration

    (4 marks)

    b) Different techniques can be applied to compose several components. Discuss **TWO (2)** types of component composition.

    (6 marks)

11. The X-model shown in **Figure 1** depicts that there are two types of repositories, namely the TCR and RCR. Discuss the usage of the TCR and RCR during the component-based development. (You may use an example to facilitate the discussion)

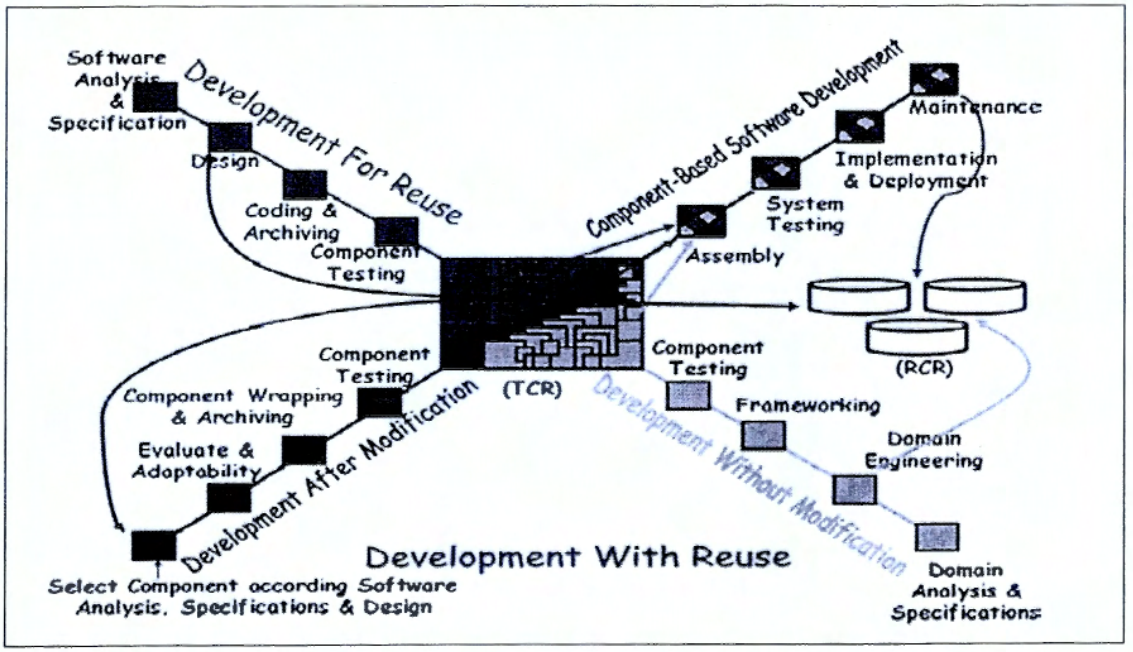


Figure 1. The X-Model

(10 marks)

END OF QUESTION